

Entrepreneurial Finance & Venture Capital

Session 1 · Lecture Notes | Prof. Ramana Nanda

1. Course Overview

Goal & Setting

This course helps **entrepreneurs and investors make better financing decisions** in early-stage, high-growth firms — settings characterised by high uncertainty, limited historical data, and skewed (power-law) outcome distributions.

- Most startups fail completely; a small number generate enormous returns — standard normal-distribution finance tools don't apply.
- Multi-stage financing is central: each round is a real option (invest more, wait, or stop).
- The VC raises money from its own limited partners (LPs), so LP incentives cascade down and shape how VCs interact with founders.

Course Philosophy

*Prof. Nanda's core message: there is rarely one 'correct' answer. What matters is **internal consistency** — your assumptions must logically lead to your conclusions. If you choose path A, your reasoning must support A, not B.*

Key Topics (across the full course)

- Valuation of startups: VC method, FCF models, decision trees
- Sources of capital: angels, VCs, venture debt, bank debt
- When / how much / from whom to raise
- Term sheets, dilution, cap tables
- Multi-stage financing and real options
- Network effects & business model impact on cash flows

2. The POCD Framework

A structured lens for evaluating any new venture (developed by Gil Salman, HBS). Each element must *fit* with the others — a good opportunity with the wrong team or deal structure leads to misaligned incentives.

P — People

Who are the founders? Relevant experience, complementary skills, prior working relationship, personal commitment. *Red flags*: first-time B2C founders with only B2B experience; founders geographically dispersed; personal distractions.

O — Opportunity	The core idea and its NPV potential. Evaluate: business model, target market segment, competitive dynamics (barriers, network effects, first-mover value), and the projected cash-flow profile.
C — Context	Macro and industry environment in which the opportunity exists. The same idea can be good or bad depending on context (regulation, technology readiness, macro conditions).
D — Deal	The financial and contractual terms: how risk and reward are allocated between founders, investors, customers, employees, and suppliers. Incentive alignment is everything.

3. Case Study: Strava (2012)

Background

Strava is a subscription-based social network for cyclists and runners. Founded by **Michael Horvath** and **Marc Gagne**. The case is set around 2011–12 when Strava is considering raising a Series A (Sigma Investors proposing \$3 million).

POCD Applied to Strava

People	Worked together before (positive). Previous venture was B2B; Strava is B2C (relevant-experience gap). Geographically split. One founder has personal distractions. They were users of their own product — strong product insight.
Opportunity	Social network for individual endurance sports (running/cycling). Sports done alone but with a competitive/social dimension — unique niche. Chose the <i>avid</i> market (not amateur, not pro): frequent users, aspirational identity, less pressure to match pro-level features.
Context	Rise of 'quantified self'; GPS and wearables proliferating; affluent cyclists willing to pay premium. Competition from Garmin, Nike etc. but network effects create defensibility if first-mover advantage can be locked in.
Deal	\$4.5M total raise; Sigma puts in \$3M for ~40% of company. Key question: is \$3M the right amount? See financing discussion below.

Network Effects & Competitive Moat

Network effects are the key source of value and defensibility:

- Each new user makes the platform more valuable for existing users (Metcalfe's Law: value $\propto n^2$).
- More users → lower customer acquisition cost (CAC) and lower churn.
- Both effects directly improve unit economics (LTV / CAC ratio).

Key insight: First-mover advantage in a network-effects business is not just about being early — it's about the lock-in that results once the network is large. Strava must grow fast to prevent rivals from establishing their own networks first.

4. Customer Lifetime Value (LTV)

Strava's Unit Economics

Revenue / customer / year	\$40
Service cost / customer / year	\$2
Annual margin / customer	\$38
Renewal rate	80% → churn rate = 20%
Customer Acquisition Cost	\$15
Claimed LTV	\$175

How LTV is Calculated (Simple Version)

The churn rate acts like a discount rate, reducing the expected stream of future margin payments. With margin M , churn rate c , and no time-value adjustment:

$$\text{LTV} = M / c - \text{CAC} = 38 / 0.20 - 15 = 190 - 15 = \$175$$

Critical Assumptions & Weaknesses

What Strava's formula ignores: there is no time-value of money discount rate. Adding a required return r to the denominator lowers LTV: $M / (c + r)$. The true LTV is almost certainly below \$175.

- No discount rate — overstates present value of future cash flows.
- Assumes current churn rate (20%) is stable as the company scales into less-avid users (churn likely rises; CAC likely rises too).
- Positive offset: growing network effects could *lower* churn over time.
- As an investor: always ask what assumptions drive the LTV, and stress-test them.

5. Cash Flow Profile & Deferred Revenue

Cash Flow ≠ Profit

This distinction is foundational. Strava reaches **cash-flow breakeven in 2012** but is not *profitable* until 2014. An investor cares about cash, not accounting profit.

- From the operator's perspective: cash flow statement is what matters; the P&L; is an accounting construct.
- Key formula: $\text{FCF} = \text{EBIT}(1 - \text{tax}) - \Delta \text{NWC} - \Delta \text{Net Fixed Assets}$.

Deferred Revenue — Why It's a Feature, Not a Bug

Strava is a subscription business: customers pay upfront. Under GAAP, revenue can only be recognised when delivered, so unearned subscriptions sit on the balance sheet as a **liability (deferred revenue)**.

- Deferred revenue = negative working capital → cash inflows exceed profit.

- This is **customer financing**: the business is funded by its own customers, reducing reliance on external capital.
- Other examples: Amazon (supplier credit cycle), Apple subscriptions, Stripe.

Counter-intuitive point: having more liabilities (deferred revenue) is good here — it means customers are financing your operations at zero cost of capital.

The Cash Flow Curve Concepts

Cash flow breakeven	The point at which operating cash flows turn positive — the company no longer depends on external finance to survive.
Cumulative breakeven	When total cash received from operations recoups all prior losses. This comes <i>after</i> cash flow breakeven.
Burn rate	Rate at which cash is consumed before breakeven. Determines runway and urgency of fundraising.
Maximum financing need	The trough of the cumulative cash flow curve — the most external capital the company will ever need.

6. Financing Decision — How Much & From Whom?

Why Not Debt?

A lender's return is capped (interest only) but the risk is unbounded downside (default). For a high-risk startup with no assets to collateralise:

- No hard assets → nothing to seize if the company fails.
- All value is in projected growth → speculative, unverifiable.
- **Conclusion:** equity (VC) is the natural financing vehicle because it allows the investor to share in the upside commensurate with the risk taken.

Angels vs. VCs — Misaligned Growth Preferences

Angel investors	Often comfortable with slower, capital-efficient growth. Less pressure to exit quickly. May not have deep pockets for follow-on rounds.
VC investors	Need large exits to generate fund-level returns (power-law payoffs). Push for aggressive growth to win the market quickly.

***Critical insight:** financing strategy is NOT separate from product-market strategy. Who funds you shapes how you grow, and your growth path determines who will fund you. These decisions must be made together.*

The \$3M Question — Counter-Intuitive Insight

Naively: Strava's projected burn is ~\$500K so why raise \$3M? But in a subscription business the logic reverses:

- **In manufacturing:** slower growth → less inventory/receivables needed → less cash consumed.

- **In subscription SaaS:** slower growth → fewer upfront subscription payments → *less* cash generated → you actually need *more* external capital.

*Therefore: raising more than you strictly need is an **insurance policy**. If you miss growth targets, you still have runway. If you nearly run out of cash, investors will know — and will extract punishing terms.*

- Cash on the balance sheet also improves your negotiating position: investors want in to a company that doesn't *need* them.

7. Key Takeaways

POCD	Evaluate People, Opportunity, Context, Deal — and check they fit together.
Network effects	Drive down CAC and churn simultaneously; create defensible moat; reinforce first-mover advantage in platform businesses.
LTV formula	$LTV = (\text{Annual margin}) / (\text{Churn rate}) - CAC$. Always challenge the missing discount rate and scaling assumptions.
Deferred revenue	Subscription businesses with upfront payments generate cash faster than profit — sometimes before they're profitable.
Cash ≠ Profit	Operate from the cash flow statement. P&L; is an accounting construct; survival depends on cash.
Subscription paradox	Growing slowly in a subscription business actually increases cash needs (opposite of manufacturing).
Financing & strategy	Who you raise from shapes your strategy. Do these together, not sequentially.

Next session (Session 2): Free Cash Flow drivers in depth — profitability (p), asset intensity (a), and growth (g) — and the 3-stage FCF valuation model.